



Technical Service Bulletin: TSB110020

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Main Bearing Bore Inspection and Repair Guidelines (T814)

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Manuals affected by this service parts topic:

- M11 Shop Manual, Bulletin Number 3666075
- L10 and M11 Alternative Repair Manual, Bulletin Number 3810310

This document supersedes the following:

- Service Parts Topic 00T1-4
- Service Parts Topic 00T1-4a
- Technical Service Bulletin TSB110283
- Technical Service Bulletin TSB120142

This service parts topic provides main bearing bore inspection and repair guidelines for the main bearing caps when there is evidence of main bearing cap fretting. These guidelines are **only** for M11/ISM and N14 Series engines.

Main bearing cap fretting is attributed to the failure of the main bearing capscrews to provide sufficient clamp load for extreme duty cycles. Fretting at the block/main bearing cap interface results in an oval shape and an overall decrease in the main bearing bore size. This decrease reduces the oil film thickness, which can lead to spun main bearings.

Reuse Guidelines

The following procedure need **only** to be performed if main bearing cap fretting is discovered during a main bearing inspection or a failure analysis of a spun main bearing.

Remove main bearing caps (no more than two at a time) to inspect for indications of fretting.

On the M11/ISM engines, fretting typically occurs on the number 2 and number 6 main bearing caps; therefore, inspection of the number 2 and number 6 main bearing caps **only** is sufficient.

On the N14 engine, fretting typically occurs on the number 2, number 3, number 5, and number 6 main bearing caps; therefore, inspection of the number 2, number 3, number 5, and number 6 main bearing caps **only** is sufficient.

If fretting on any of the main bearing caps is occurring, the inspection procedure, detailed in this topic, **must** be performed. Before performing the bore inspection, determine if the bearings are reusable according to the guidelines in the appropriate troubleshooting and repair manual. Also determine if there is side-to-side press fit of the main cap in the block. If there is, proceed with the inspection; if there is **not** a side-to-side press fit, a service cap **must** be used. If a service cap is required, the inspection procedure is **not** required and the engine will need to be removed from the chassis and prepared for line boring.

- For clearance greater than or equal to 0.051 mm [0.002 in], reassemble the main caps according to the guidelines presented in this topic. On the M11/ISM engines, the Torque-Plus-Angle (70 ft-lb plus 120-degree) method **must** be used for all engines. On the N14 engine, use the torque specification in use at the time the engine was built.
- For clearance less than 0.051 mm [0.002 in] on any main bore (or without a press fit) the engine needs to be removed and prepared for line boring. (Line-boring guidelines are included in this topic.)

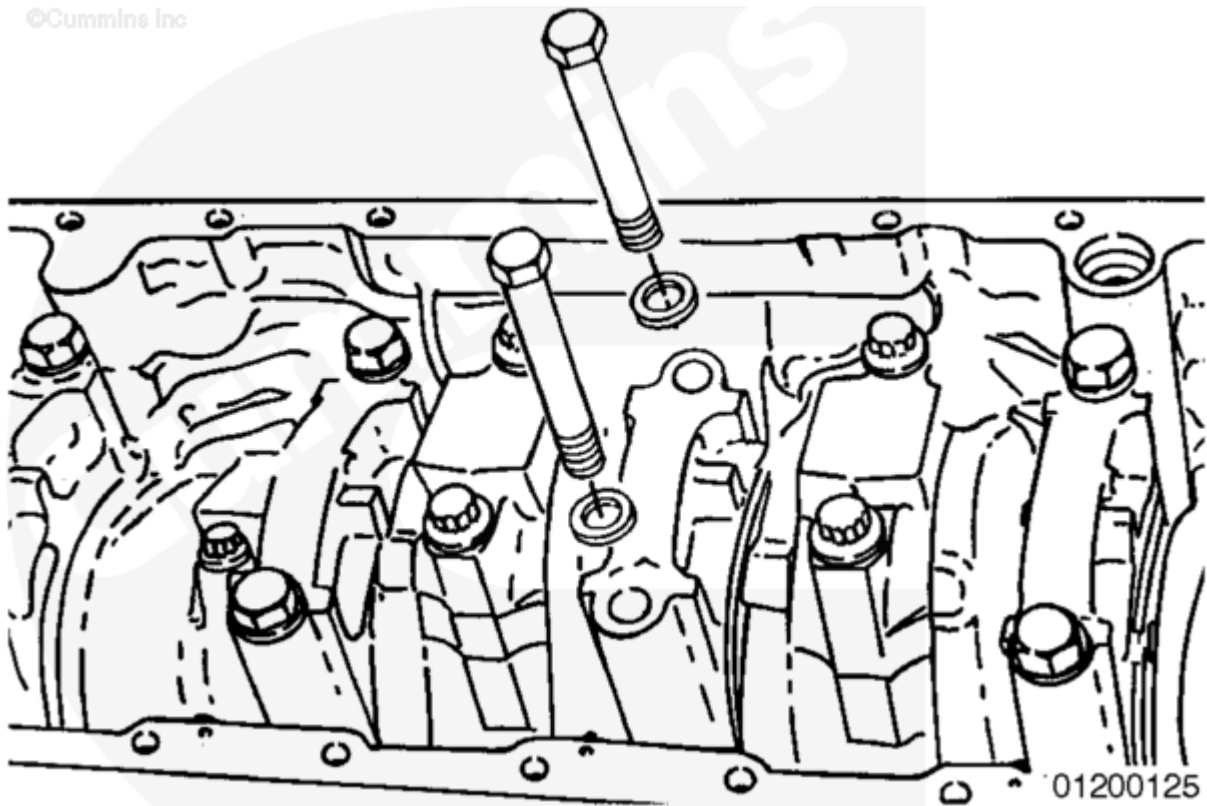
Plastigage™

Plastigage™ provides a fast and accurate method to check bearing clearances. Plastigage™ is a special extruded plastic thread of a definite diameter with accurately controlled crush properties.

Plastigage™ is packaged in calibrated envelopes, 12 envelopes per box. These envelopes not only protect the plastic threads but also serve directly as scales to measure the bearing clearance. Both sides of the envelope have a printed scale of graduations. One side is calibrated in inches, the other in millimeters. The numbers on the scale are the bearing clearances in thousandths of an inch or millimeter. When the width of a compressed section of Plastigage™ in a bearing or journal is compared with the appropriate-numbered graduation, the bearing clearance can be read directly from the scale.

Plastigage™ is available, from a local parts store, in four styles to cover different clearance ranges. Both Plastigage™ thread and its matching envelope have a distinctive color for each clearance range.

The PG-1 (green) style, clearance range of 0.025 to 0.076 mm [0.001 to 0.003 in], is required for this procedure.



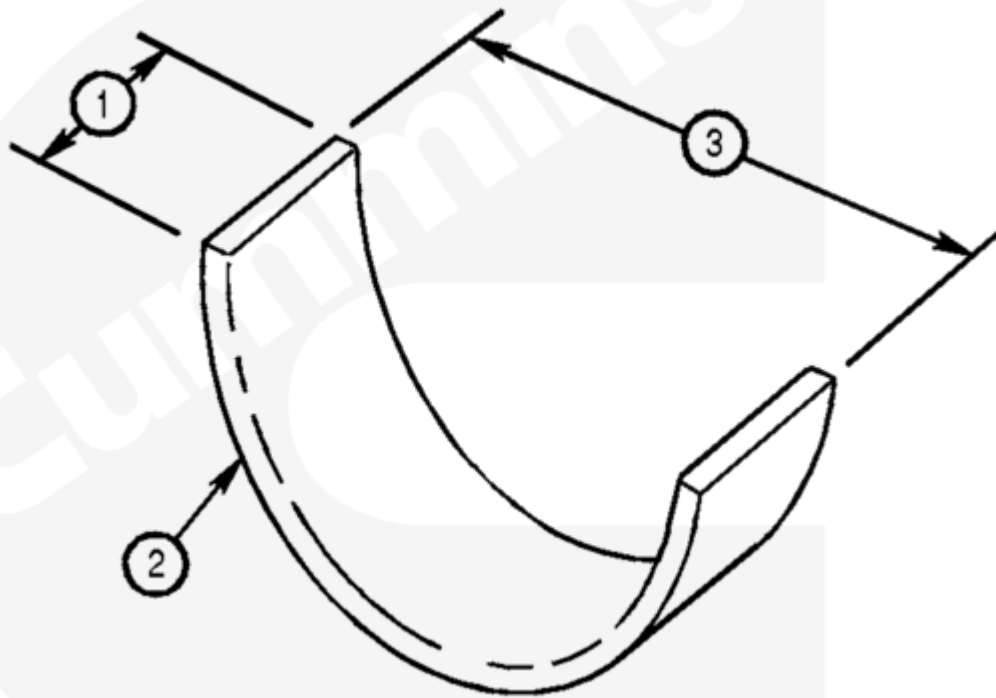
Note : Two main bearings caps must be installed at all times to support the engine crankshaft.

To check the number 2 main bearing, remove the main bearing capscrews on number 1 and number 3 main bearing caps.

Remove the number 1 and number 3 main bearing caps.

Use main bearing cap puller, Part Number ST-1178, to remove the caps. The main bearing cap puller **must** be centered on the cap. Pull straight down to remove the cap.

Remove the lower bearings.



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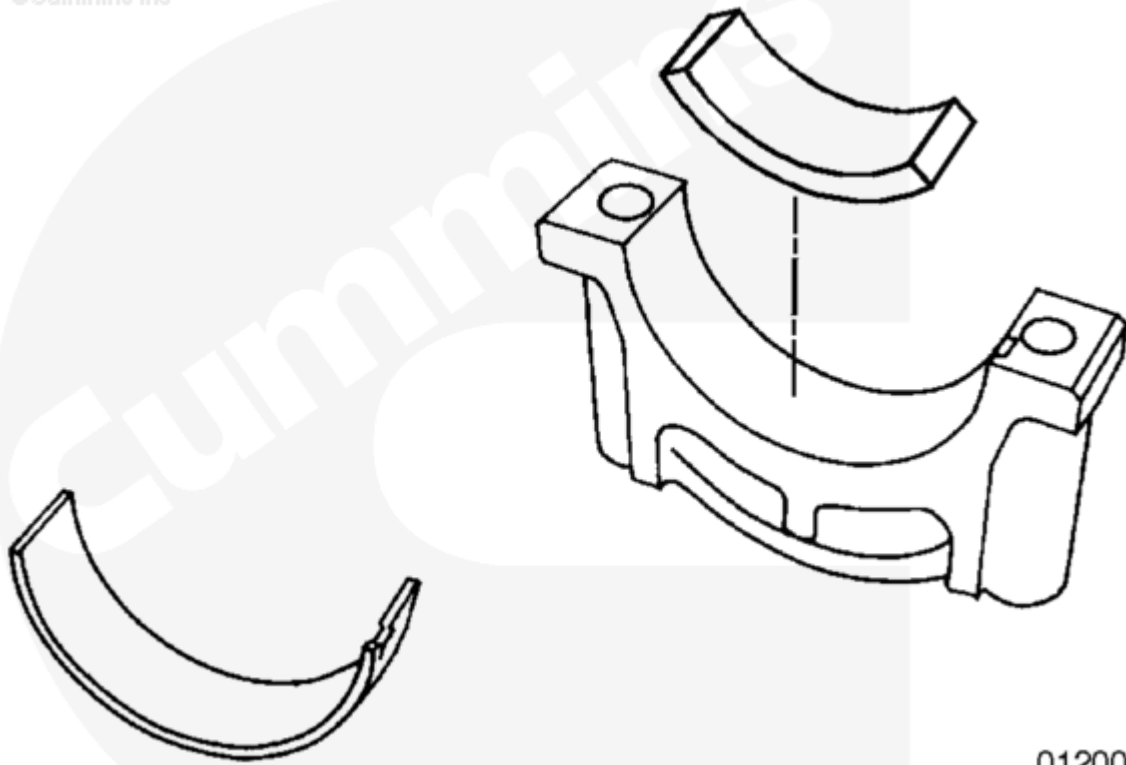
To inspect the main bearing-to-journal clearance on an in-chassis engine, shims **must** be used.

Note : The shims push the crankshaft up against the adjoining upper bearings so an accurate clearance measurement can be made on the lower bearing.

The shims can be manufactured easily from a piece of 4-inch-diameter, schedule 40 PVC pipe. The dimensions of the fabricated shims **must** be as follows:

1. Width = 1.5 inches
2. Thickness = 0.25 inch (approximate thickness of schedule 40 PVC pipe)
3. Diameter = 4 inches

Note : Cut the pipe in half. The shims **must** be fabricated to the above-specified dimensions and be less than 180 degrees in circumference.



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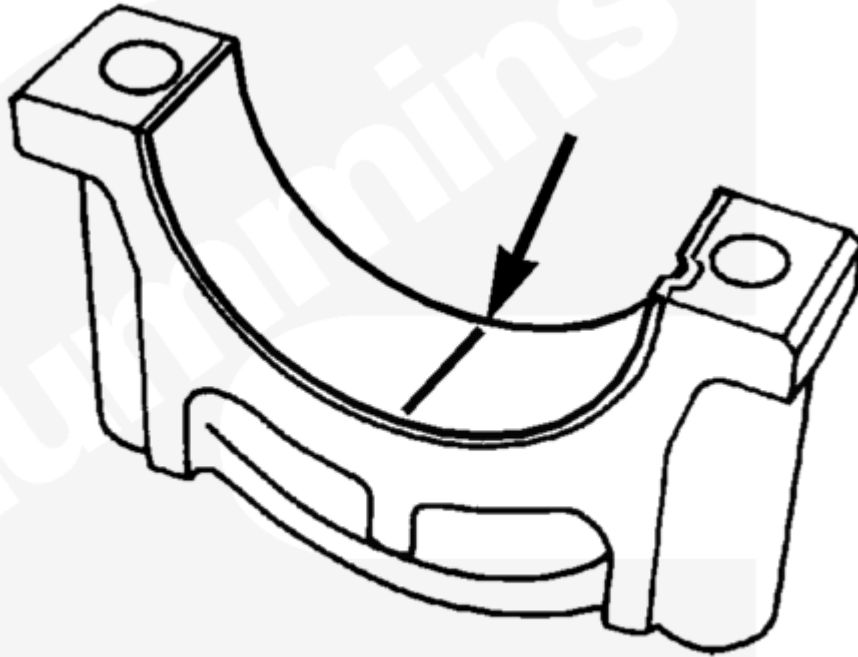
Note : Two fabricated shims are required for this procedure.

Install the shims in the main bearing caps on both sides of the bore being checked. (For example, to check the number 4 main bearing cap, install the shims on the number 3 and number 5 main bearing caps. To check the number 6 main bearing cap, install the shims on the number 5 and number 7 main bearing caps.)

Install the caps

Install and tighten the capscrews.

Torque Value: 34 N•m [25 ft-lb]



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Remove the number 2 main bearing cap.

Wipe the oil from the bearing and crankshaft journal.

Plastigage™ as long as the full bearing width. **Do not squeeze the gauge material.** (Tear through both the envelope and plastic thread at the same time.) Open the envelope lengthwise by cutting the edge with scissors or a knife or by tearing. With the envelope open, lift out the Plastigage™ by one end. (Roll out with the thumb if the material is stuck lightly to the paper.) Occasionally, the color in the Plastigage™ material will transfer to the paper. This has no effect on accuracy.

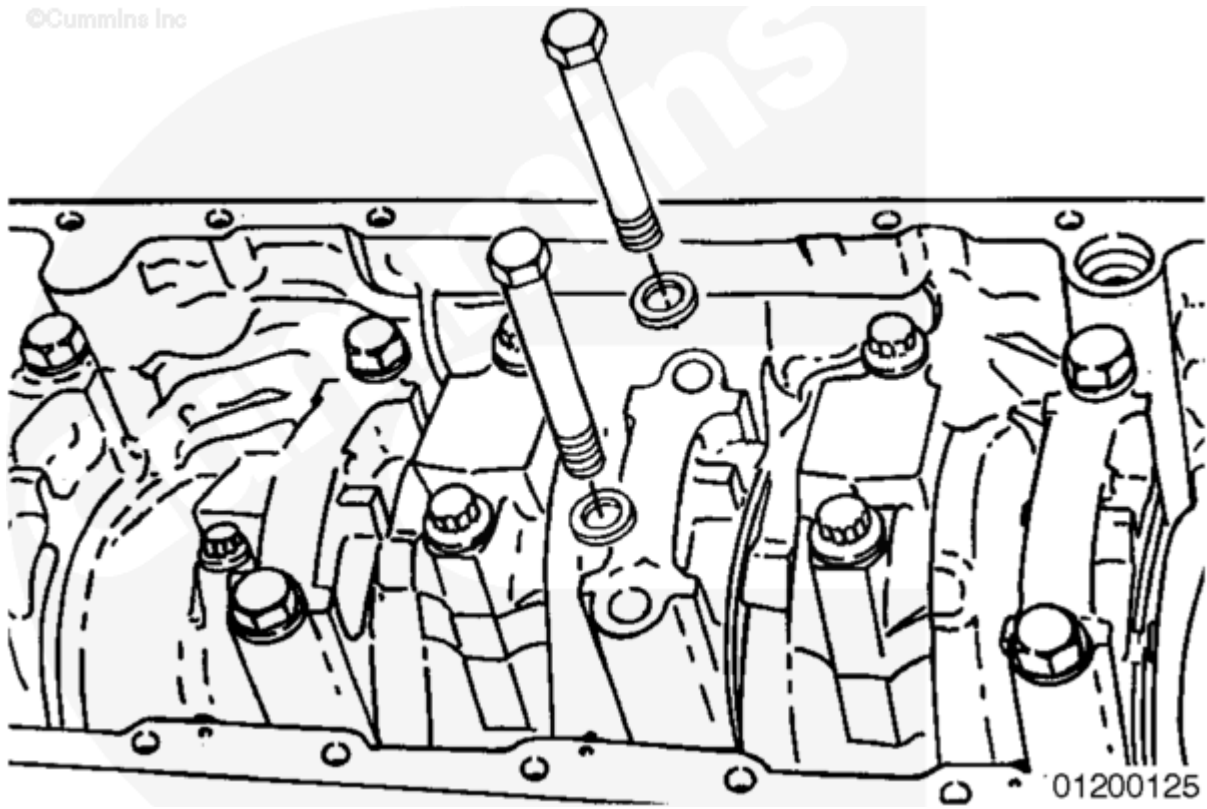
Place the piece of Plastigage™ the full length of the bearing. Insert approximately ¼ inch off center. (See the accompanying illustration.)

Note : The Plastigage™ clearance check procedure **must** be performed with the bearings that will be used in the engine.

Note : The minimum acceptable clearance is 0.051 mm [0.002 in] for both the M11/ISM and N14 Series engines.

If fretting in any degree is present but the bearings are acceptable for reuse, check the bearing thickness according to the appropriate troubleshooting and repair manual. If the bearings are reusable, perform the Plastigage™ clearance check procedure.

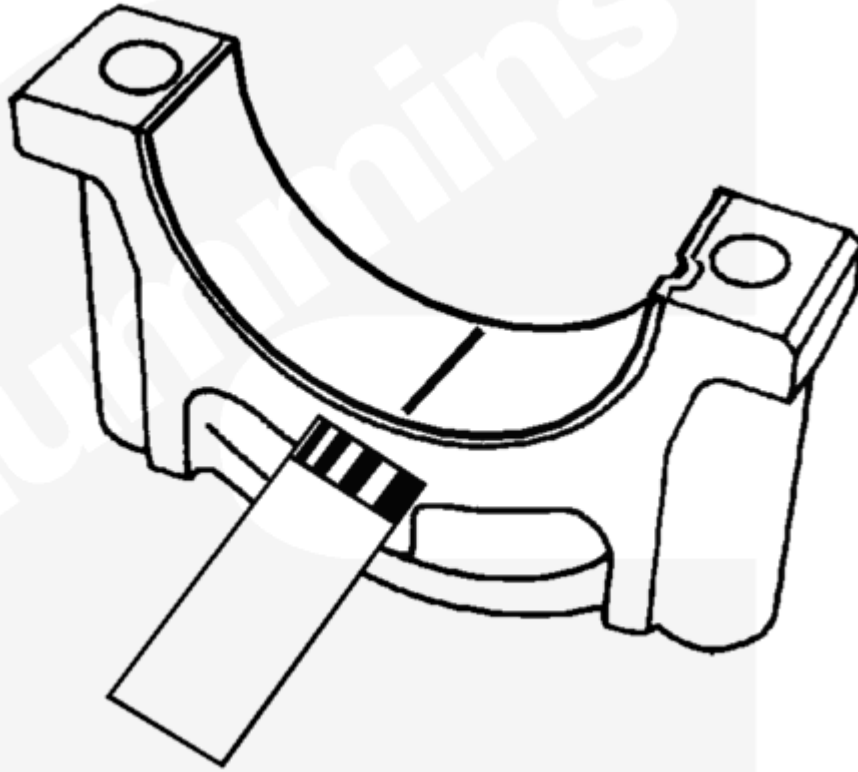
If fretting in any degree is present and the bearings appear to be damaged or worn beyond reuse, the Plastigage™ clearance check procedure **must** be performed with a new main bearing set.



Install the number 2 main bearing cap using the appropriate torque procedure. The torque procedures are located later in this service parts topic.

⚠ CAUTION ⚠

Do not disturb the Plastigage™ before reading the measurement. Disturbing the Plastigage™ will result in an incorrect measurement.



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Remove the main bearing cap. The compressed Plastigage™ will be located adhering to either the bearing shell or the crankshaft.

Compare the width of the compressed Plastigage™ at its widest point to the graduations on the envelope. The number within the graduation on the envelope indicates the bearing clearance in thousandths of an inch or millimeter, depending on which side of the envelope is used.

If the compressed Plastigage™ is less than the graduation marked 0.051 mm [0.002 in] at its widest point the block **must** be pulled from service and either line-bored or replaced.

If the clearance is greater than 0.051 mm [0.002 in] (**must** also have a press fit), reassemble the main bearing caps with existing or new main bearings.

If the clearance is less than 0.051 mm [0.002 in] on any main bearing bore (or without a side-to-side main cap press fit), the engine **must** be removed and prepared for line boring. (The line-boring procedure can be found at the end of this topic.)

Repeat the above steps to check the clearance at the number 6 main bearing cap (M11/ISM Series engine **only**)

Repeat the above steps to check the clearance at the number 3, number 5, and number 6 main bearing caps (N14 Series engine **only**)

Install (M11/ISM Series Engines Only)

Application of Loctite® 620 to Main Bearing Cap Surfaces

Note : If fretting is visible on main bearing cap surfaces number 2 and number 6 but the Plastigage™ clearance check procedure shows that the fretting is acceptable, apply Loctite® 620 to all of the main bearing cap surfaces as described in the following procedures.

Loctite® 620 adds an additional 70 percent to the shear strength of the main bearing cap joint.

The following procedures also require the use of Loctite® Primer-N.

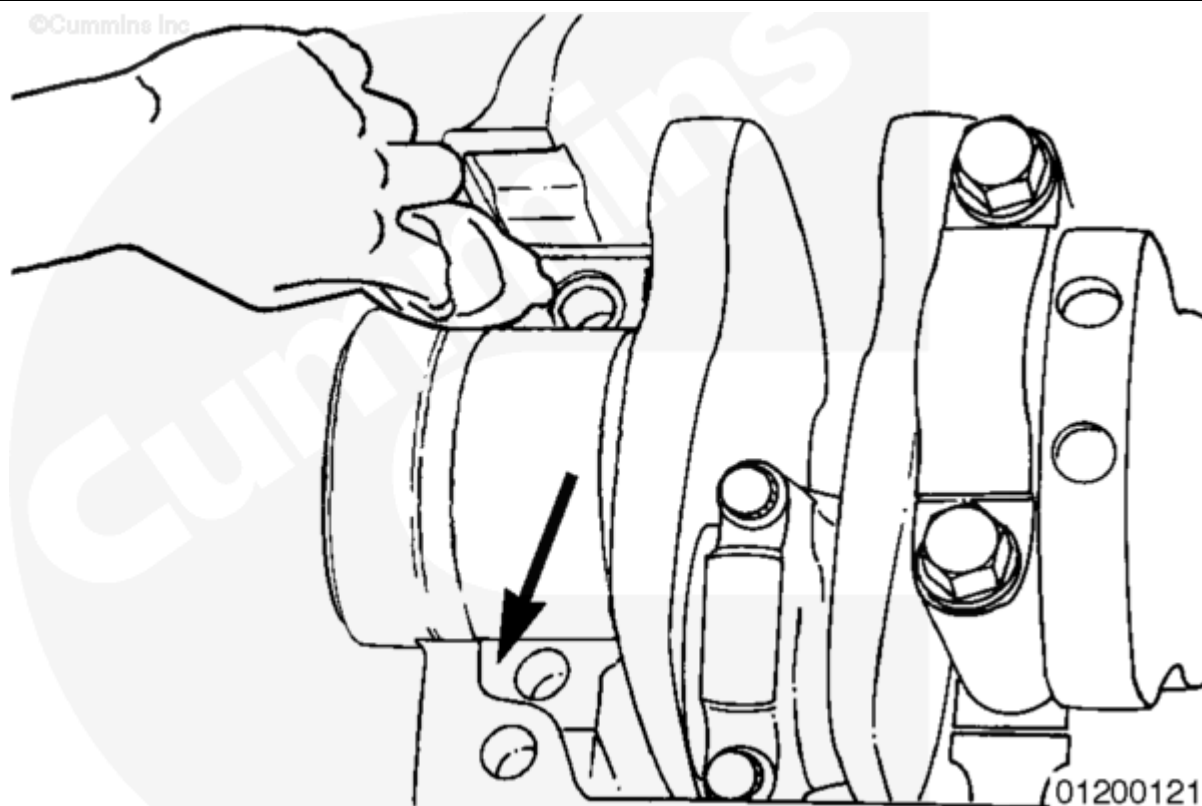
Loctite® 620 and Loctite® Primer-N are available at local parts stores. There is **no** Cummins® part number available.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.



Clean all main bearing cap mounting surfaces on the cylinder block with a solvent-based degreaser.

Dry the main bearing cap mounting surfaces on the cylinder block with a clean, lint-free cloth.

Note : The main bearing cap surfaces must be free of grease or oily residue.

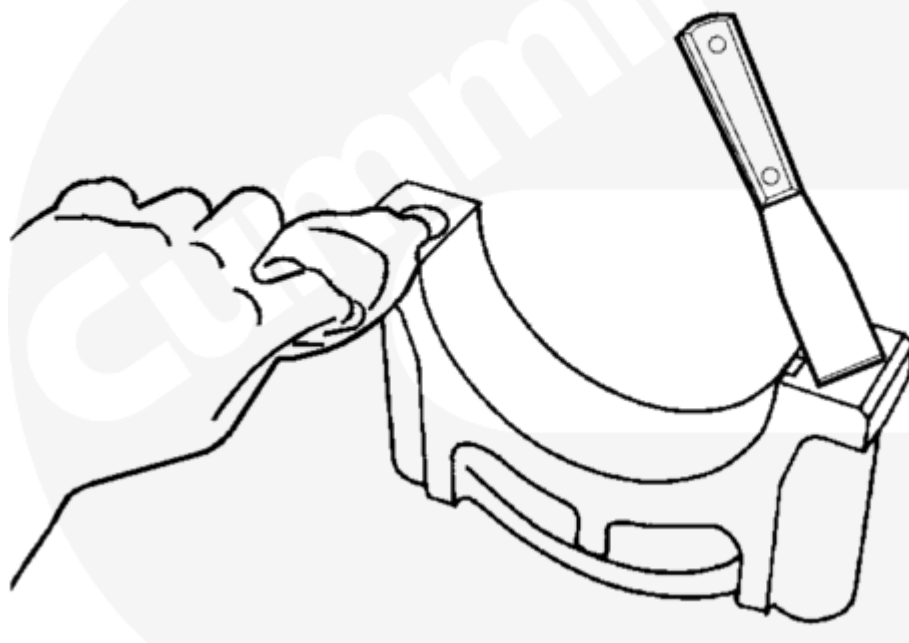
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

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Clean the mounting surfaces of the main bearing caps with a solvent-based degreaser.

Dry the mounting surfaces of the main bearing caps with a clean, lint-free cloth.

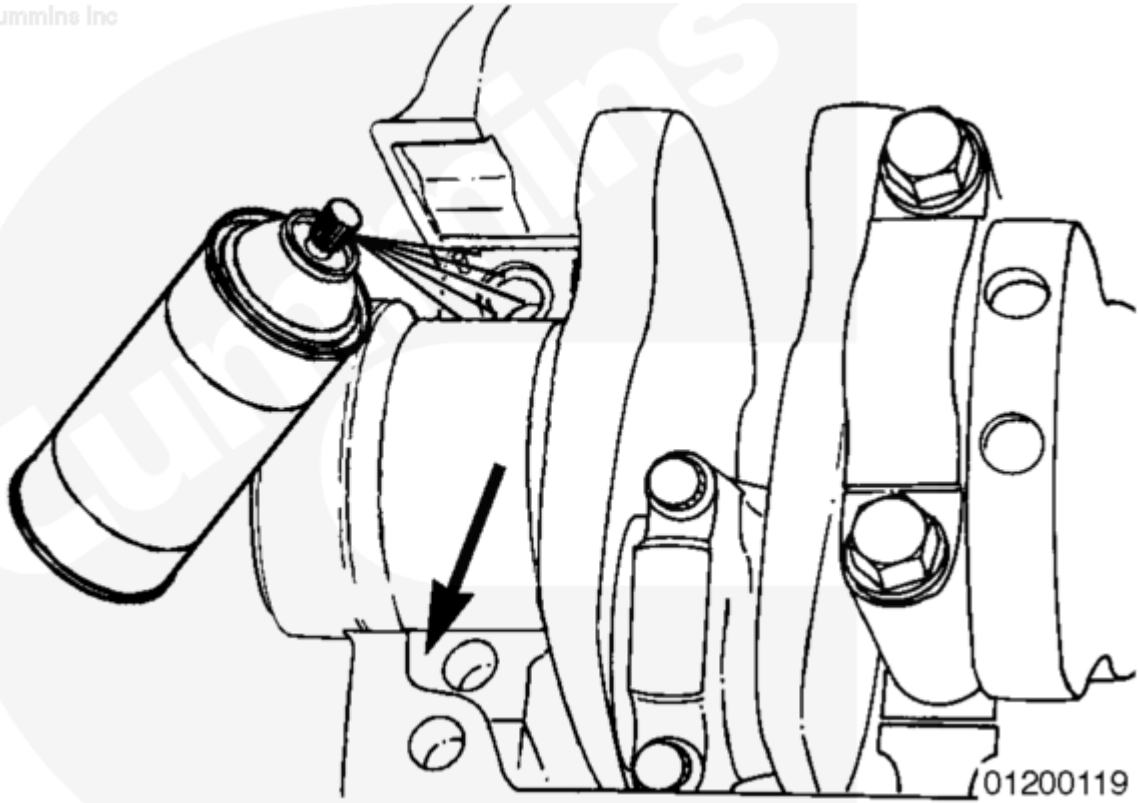
Note : The mounting surfaces of the main bearing caps must be free of grease or oily residue.

⚠ WARNING ⚠

Wear personal protective equipment when spraying Loctite® Primer-N to avoid personal injury.

⚠ CAUTION ⚠

To prevent oil seepage down onto the mating faces of the block mains, cleaning of the block main capscrew threaded holes is critical. (Use cotton swabs, or equivalent, to absorb as much oil as possible prior to spraying on the primer.)



Note : Mask the crankshaft mains prior to spraying with the primer.

Apply a light coat of Loctite® Primer-N to all main bearing cap mounting surfaces on the engine block.

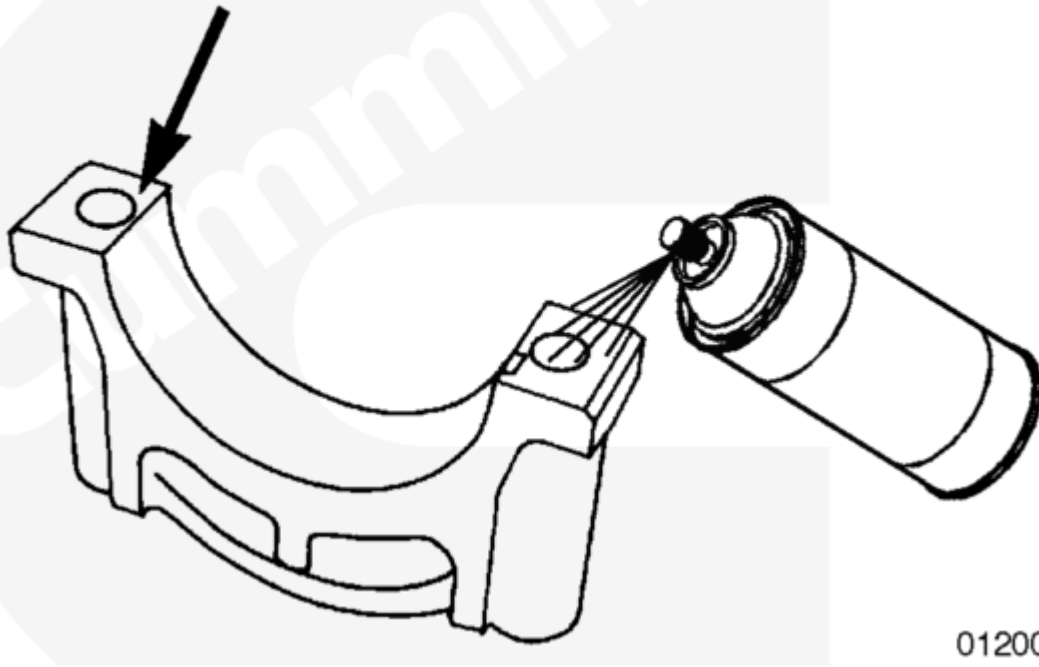
Allow to dry for 5 minutes.

⚠ WARNING ⚠

Wear personal protective equipment when spraying Loctite® Primer-N to avoid personal injury.

⚠ CAUTION ⚠

To prevent oil seepage down onto the mating faces of the block mains, cleaning of the block main capscrew threaded holes is critical. (Use cotton swabs, or equivalent, to absorb as much oil as possible prior to spraying on the primer.)



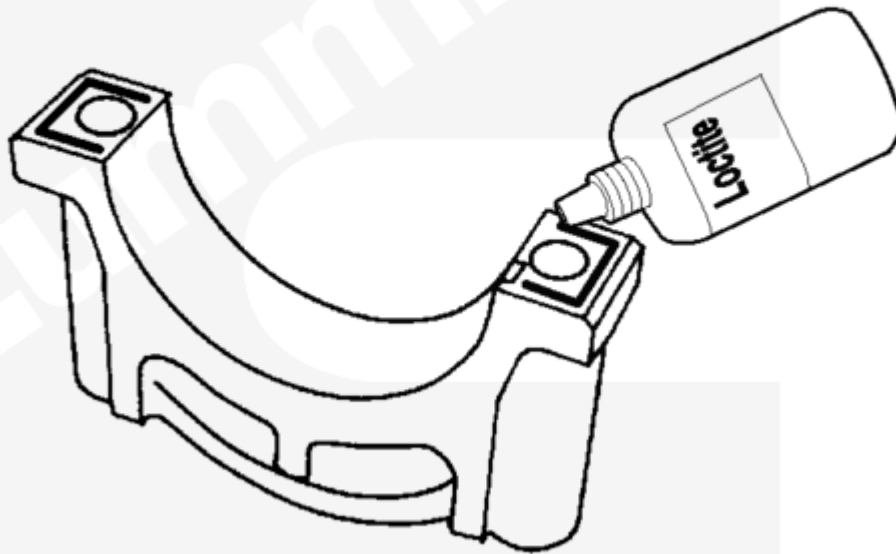
Note : Mask the crankshaft mains prior to spraying with the primer.

Apply a light coat of Loctite® Primer-N to all the mounting surfaces of the main bearing caps.

Allow to dry for 5 minutes.

⚠ CAUTION ⚠

The bead must be 3 to 5 mm [0.118 to 0.197 in] wide and must not enter the main bearing shell inside diameter.

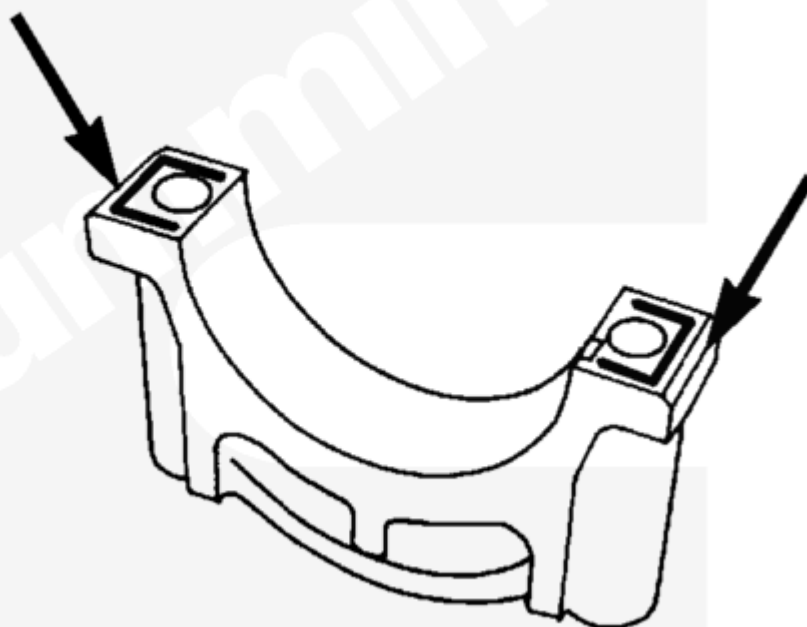


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Note : The Loctite® 620 is applied to four locations on each main bearing cap.

Apply a bead of Loctite® 620 to the main bearing cap mounting surface (as shown in the accompanying illustration).

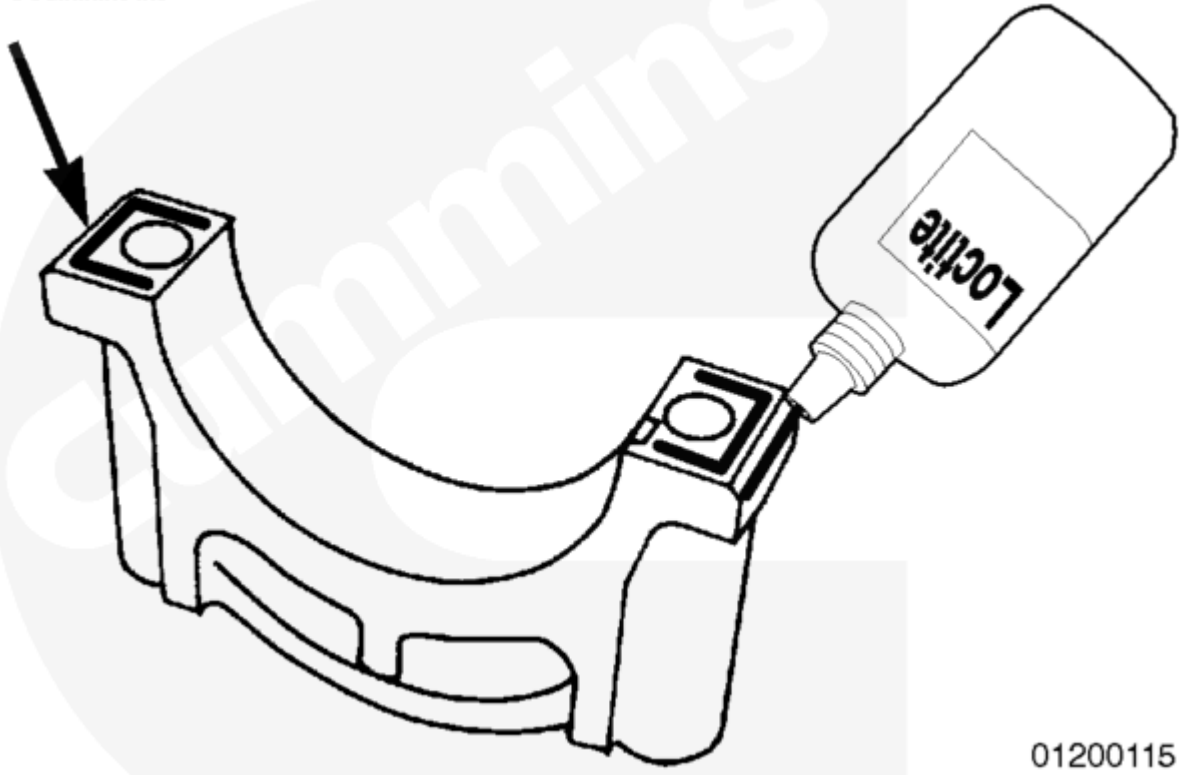
Repeat this procedure on the opposite side of the main bearing cap.



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Locate the leading edge of the main bearing cap clearance chamfer.

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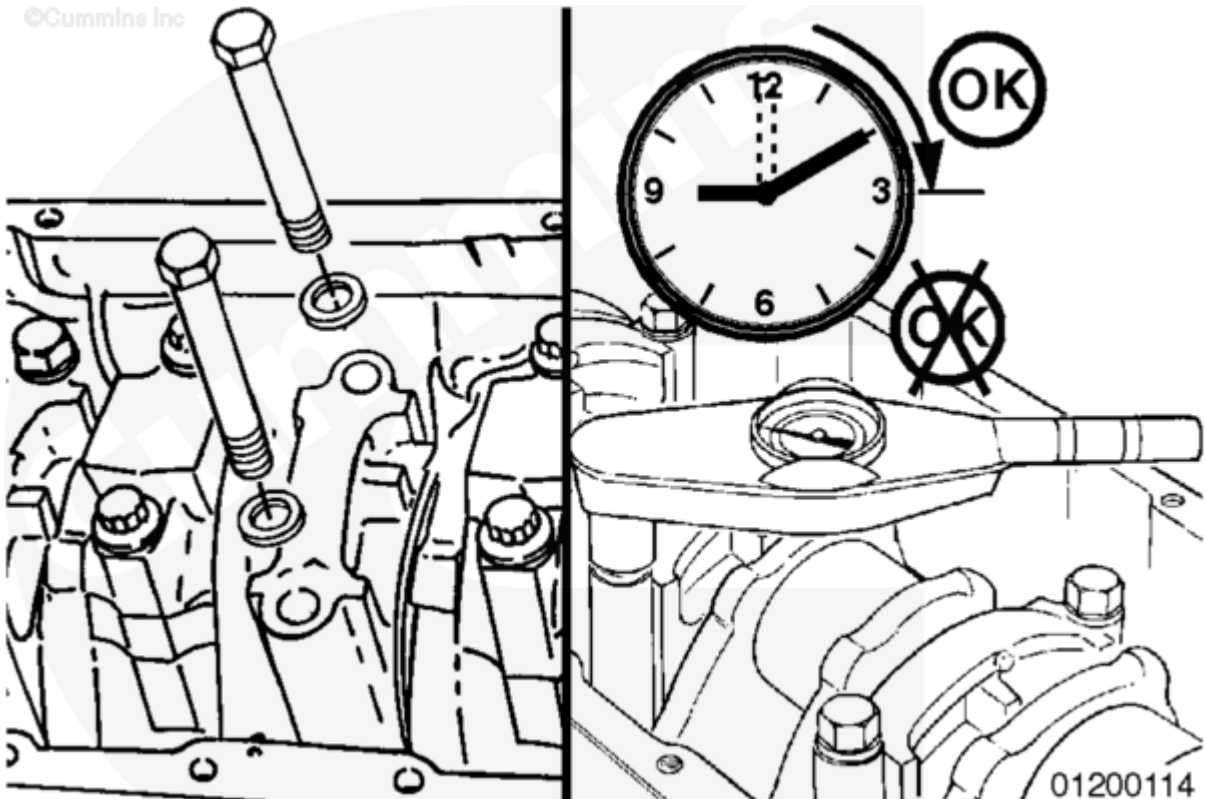


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Apply a bead of Loctite® 620 to the leading edge of the main bearing cap clearance chamfer (as shown in the accompanying illustration).

Repeat this procedure on the opposite side of the main bearing cap.

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Assemble the main bearing caps as defined in the M11 Shop Manual, Bulletin 3666075, or the Troubleshooting and Repair Manual, ISM/QSM11 Series Engines, Bulletin 3666322.

Note : To prevent possible bore size/clearance issues, the main bearing capscrews **must** be tightened within 15 minutes of the Loctite® application.

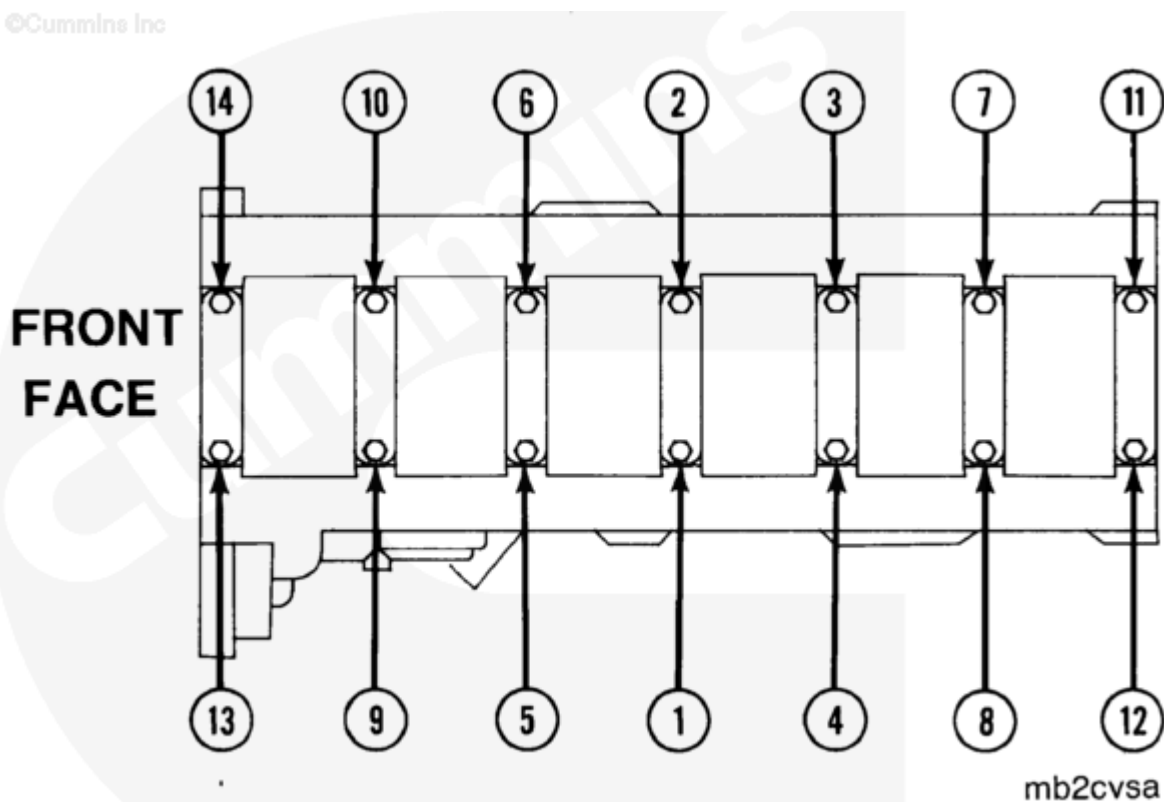
Tighten the main bearing capscrews using the torque-plus-angle method found following this procedure.

Allows the Loctite® 620 to cure fo a minimum of 3 hours before filling the crankcase with oil and running the engine.

M11/ISM Main Bearing Capscrew Torque Procedures

The torque procedure for the main bearing capscrews has changed for the M11/ISM Series engines. A torque-plus-angle method **must** be used. This method results in a more consistent loading of the joint than the previous torque-**only** method.

The new, proper torque procedure is as follows:



M11/ISM Torque Procedure

Coat the capscrew threads and both sides of the washers with clean lubricating engine oil. Drain the excess oil from the capscrews before installing them in the cylinder block. This allows the highest possible preload from the input torque.

Tighten the main bearing capscrews on the M11/ISM Series engine in the alternating sequence shown in the illustration to the following torque values:

- Step 1 **Torque Value:** 95 N•m [70 ft-lb]

- Step 2 Loosen Completely
- Step 3 **Torque Value:** 95 N•m [70 ft-lb]
- Step 4 Rotate 120 degrees
 - Use a $\frac{3}{4}$ inch drive socket that has been etched with two axial lines 120 degrees apart.
 - Place the socket over the main cap capscrew head, and mark the cap with a silver pencil at the location of the socket axial line number 1.
 - Using an impact wrench, rotate the main bearing capscrew in the direction of tightening until the socket axial line number 2 lines up with the mark on the cap (120 degrees.)

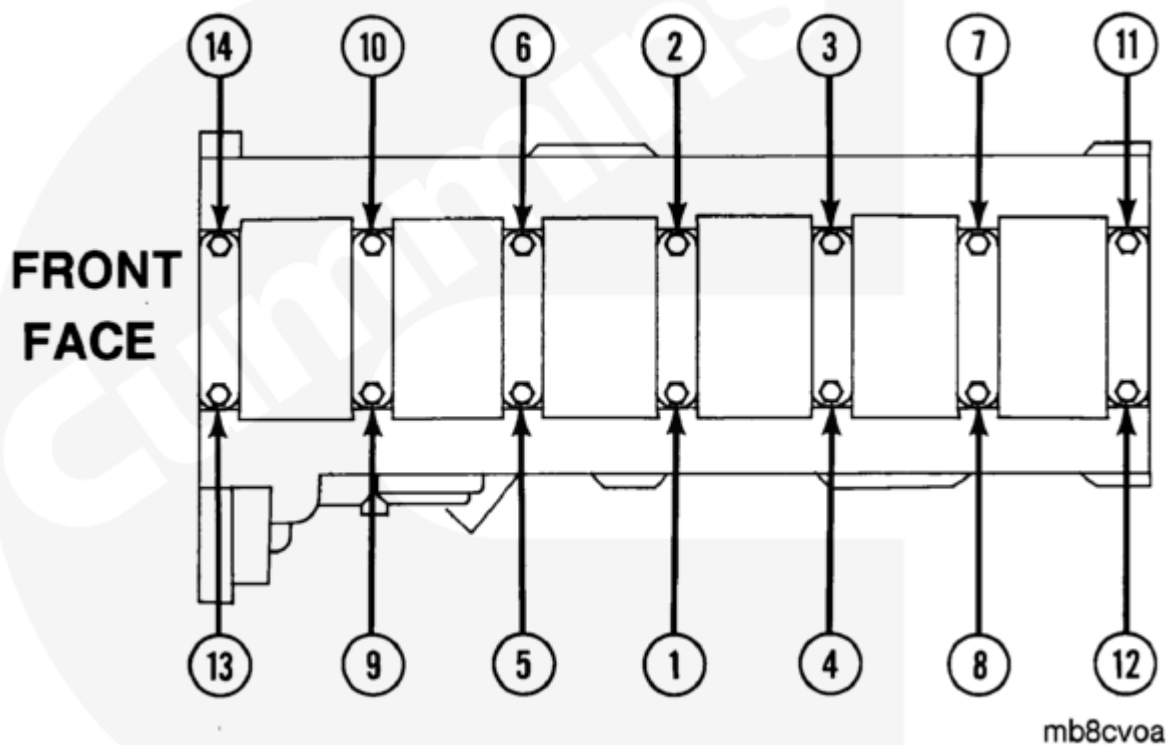
N14 Main Bearing Capscrew Torque Procedures

Depending on engine build date, there are two different procedures. They are as follows:

For engines built prior to January 5, 1998 (and prior to engine serial number 11875596), use this procedure that includes use of the $\frac{3}{4}$ inch diameter, grade 8 capscrew, Part Number 208346.

Note : Use the torque procedure in the N14 Shop Manual, Bulletin 3810487, to tighten the N14 main caps in engines that were made prior to January 5, 1998.

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Coat the capscrew threads and both sides of the washers with clean lubricating engine oil. Drain the excess oil from the capscrews before installing them in the cylinder block. This allows the highest possible preload from the input torque.

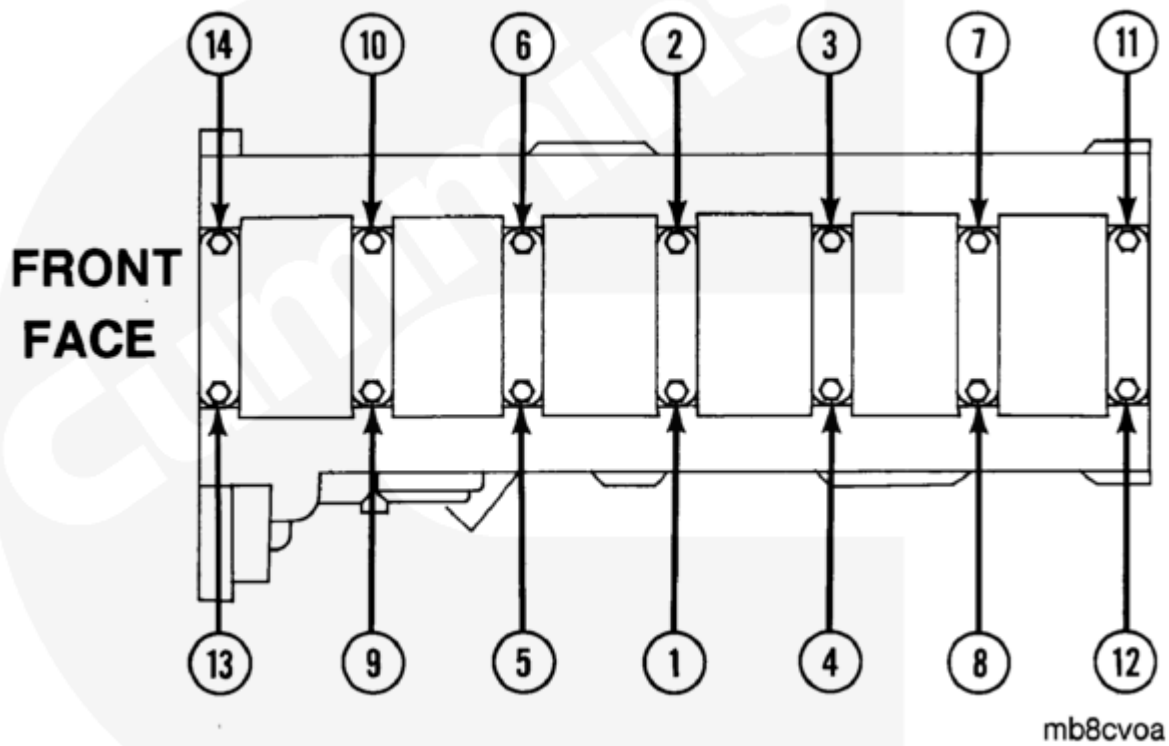
Tighten the main bearing capscrews on the N14 Series engine in the alternating sequence shown in the illustration to the following torque values:

- Step 1 **Torque Value:** 122 N•m [90 ft-lb]
- Step 2 **Torque Value:** 230 N•m [170 ft-lb]

- Step 3 **Torque Value:** 346 N•m [255 ft-lb]
- Step 4 Loosen Completely
- Step 5 Repeat Steps 1 through 3

For engines built on or after January 5, 1998, beginning with engine serial number first 11875596, use this procedure that includes a new 3/4 inch diameter, grade 12.9 capscrew, Part Number 3411337.

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Coat the capscrew threads and both sides of the washers with clean lubricating engine oil. Drain the excess oil from the capscrews before installing them in the cylinder block. This allows the highest possible preload from the input torque.

Tighten the main bearing capscrews to the following torque values:

- Step 1 **Torque Value:** 69 N•m [50 ft-lb]
- Step 2 **Torque Value:** 136 N•m [100 ft-lb]
- Step 3 **Torque Value:** 190 N•m [140 ft-lb]
- Step 4 Rotate the main bearing capscrew 90 degrees in the direction of tightening while following the instructions below:
 - Use a 3/4-inch drive 1-1/8 inch socket that has been etched with two axial lines 90 degrees apart.
 - Place the socket over the main cap capscrew head, and mark the cap with a silver pencil at the location of the socket axial line number1.
 - Using an impact wrench, rotate the main bearing capscrew in the direction of tightening until the socket axial line number2 lines up with the mark on the cap (90 degrees).
- Step 5 Loosen the main bearing capscrews completely
- Step 6 Repeat Steps 1 through 4

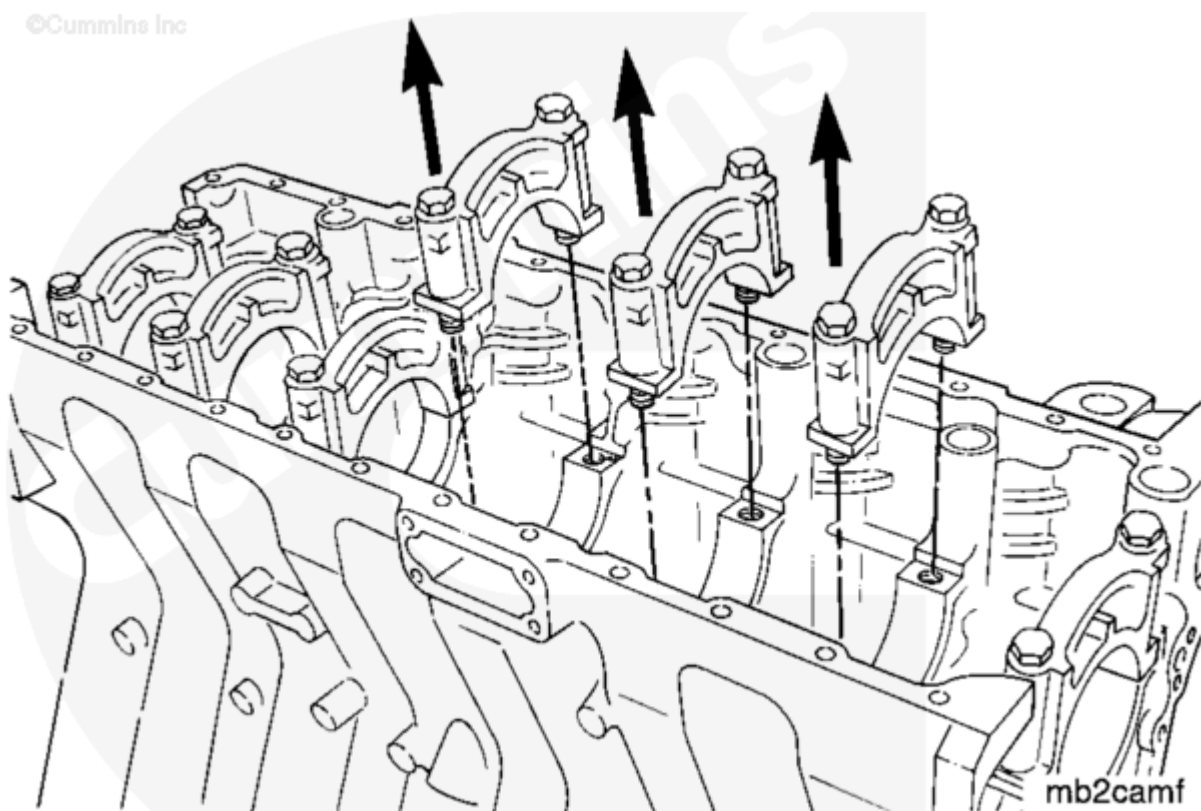
Line Bore Guidelines

M11/ISM Guidelines

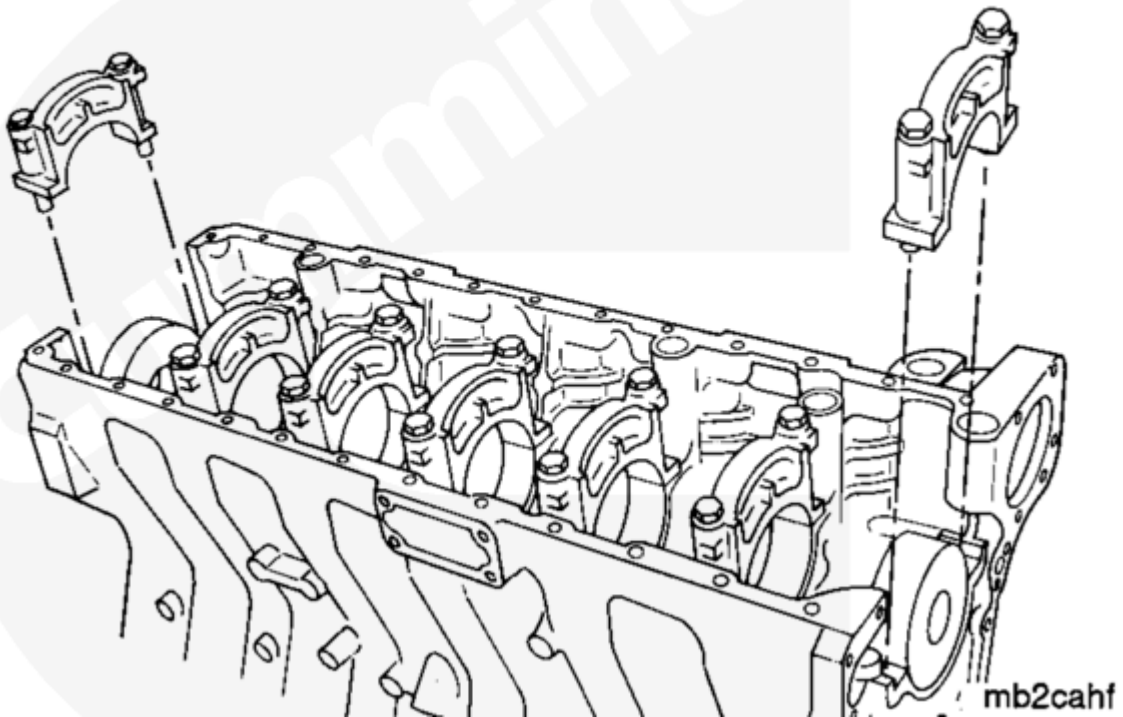
If one or more of the main bearing bores has a clearance problem (less than 0.051 mm [0.002 in]), repair of the engine requires line boring of the main bearing bore in the cylinder block. Refer to the L10/M11 Alternative Repair Manual, Bulletin 3810310-01, for the line-boring procedure.

N14 Guidelines

When it has been determined that one or more of the main bores has a clearance problem (less than 0.051 mm [0.002 in]), repair of the engine requires line boring of the cylinder block described below. Refer to Section ARM (001-020) in the Alternative Repair Manual, Bulletin 3379035.

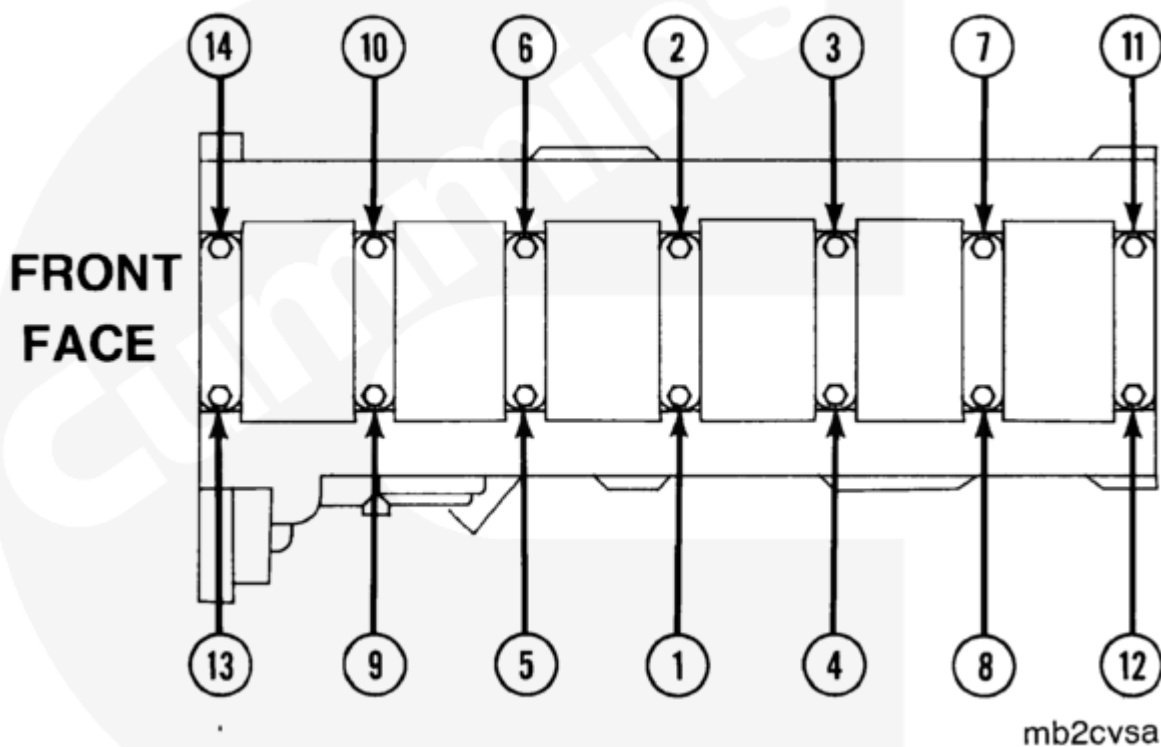


Remove the main bearings from all journals



Install two of the original main bearing caps that have no fretting with the original grade 8 capscrews.

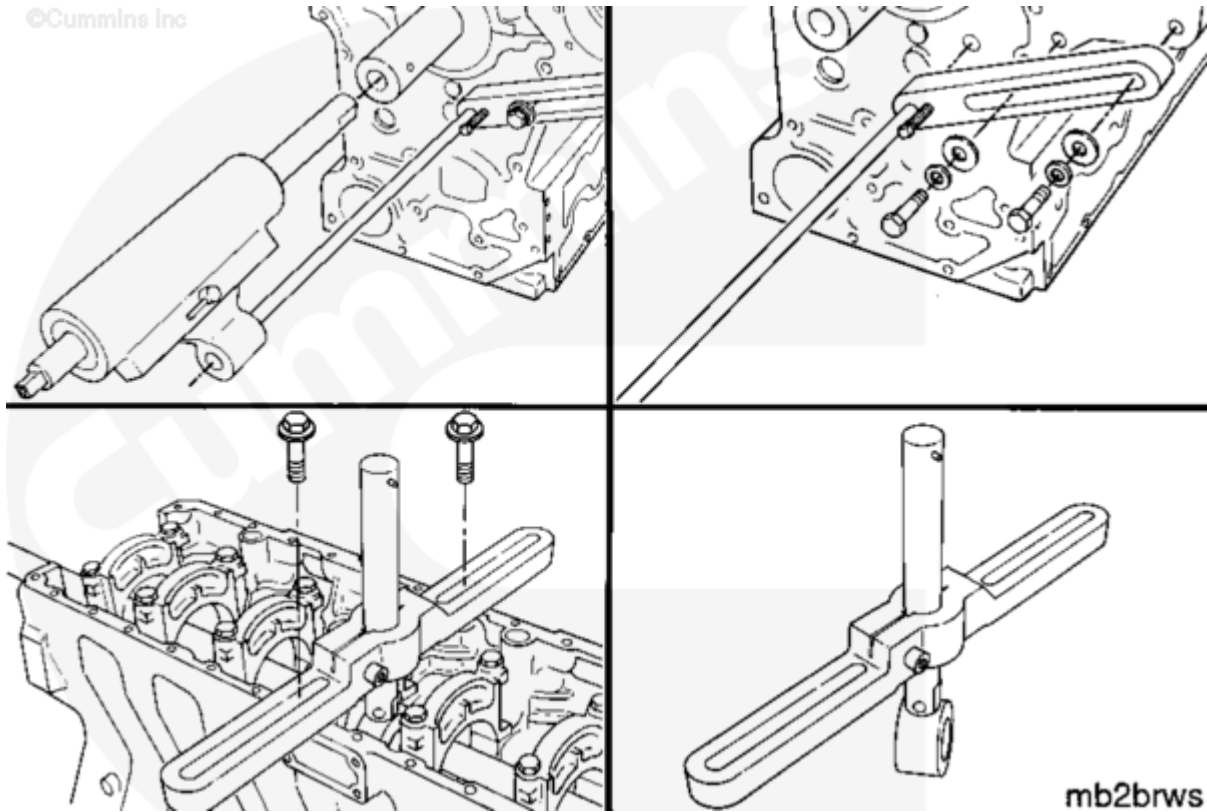
Note : It is recommended to go this in two journals that are far apart from each other (number 1 and number 7 are ideal). These journals are your reference journals.



Install the remaining main bearing caps with new grade 12.9 capscrews, Part Number 3411337.

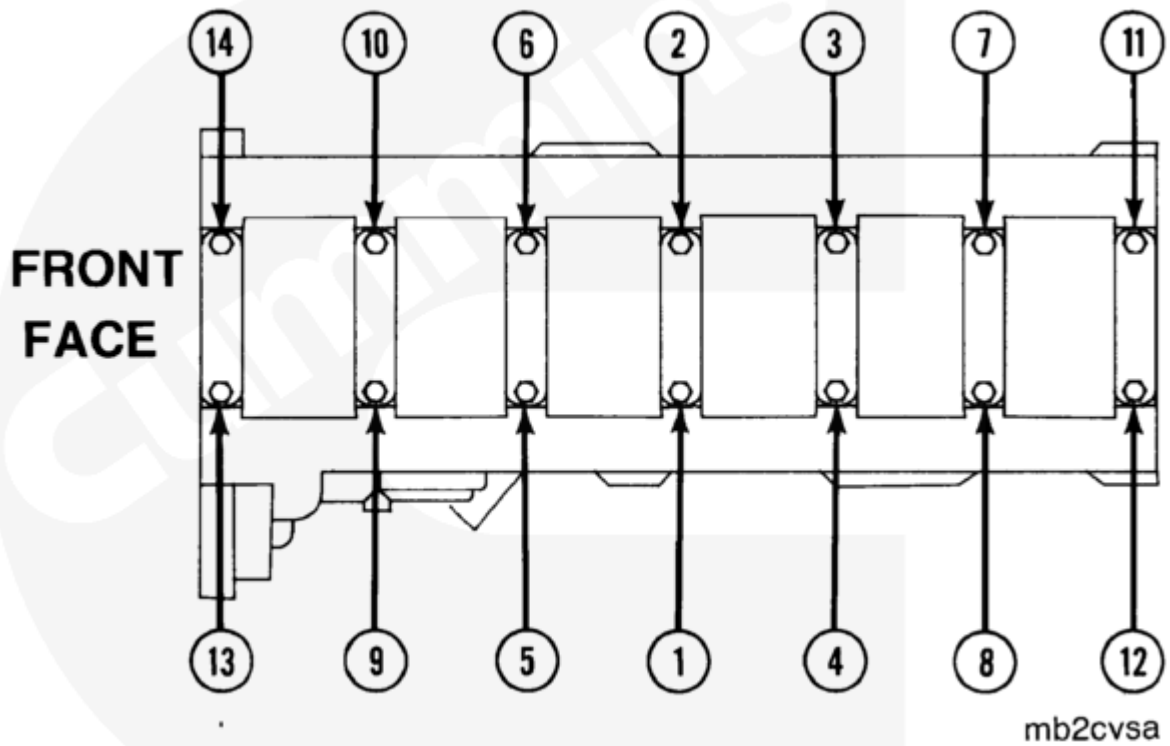
Tighten as follows:

- Step 1 **Torque Value:** 68 N•m [50 ft-lb]
- Step 2 **Torque Value:** 136 N•m [100 ft-lb]
- Step 3 **Torque Value:** 190 N•m [140 ft-lb]
- Step 4 Tighten 90 degrees

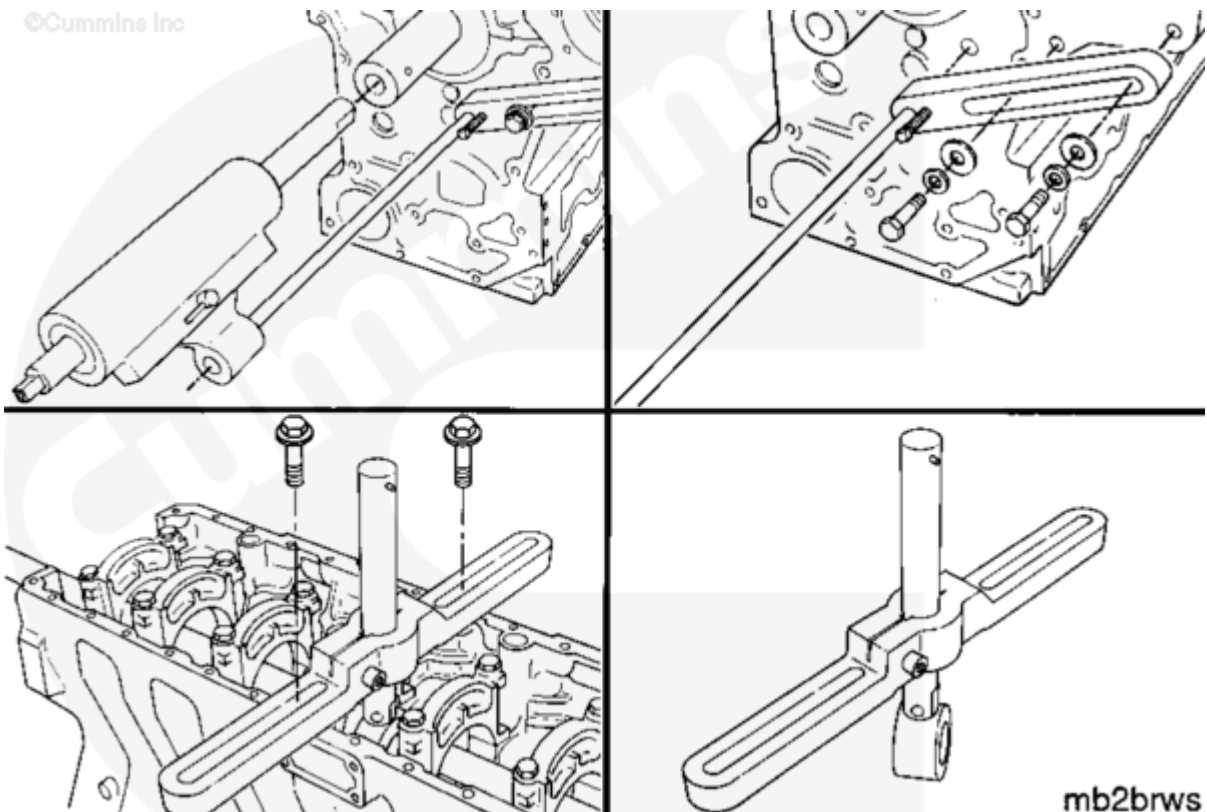


Use a dial indicator to align the boring bar with the centers of the journals that have the original grade 8 capscrews.

Line-bore the journals that have the grade 12.9 capscrews.



Replace the grade 8 capscrews in the two reference journals with grade 12.9 capscrews and tighten in accordance with the procedure outlined above.



Use a dial indicator to align the boring bar with the centers of two of the newly bored journals. Line-bore the two reference journals.

Document History

Date	Details
2011-1-14	Module Created
2015-3-27	Updated part specifications and formatting.

Last Modified: 01-Apr-2015
